VOLUME FORECASTING TECHNIQUES

Daily Volumes

Our daily volume forecasting approach is based on an autoregressive moving average (ARMA) technique. Our research finds daily volumes to be dependent upon: (1) either a moving average (ADV) or a moving median (MDV) daily volume, (2) a historical look-back period of 10 days, (3) a day of week effect, or (4) a lagged daily volume term. Additional adjustments can also be made to the volume forecasts on special event days such as earnings, index reconstitution, triple and quadruple witching days, fed day, etc. (see Chapter 2, Market Microstructure).

Our daily volume forecasting analysis is as follows:

Definitions

Historical Look-Back Period. The number of days (data points) to use in the forecasts. For example, should the measure be based on 66 days, 30 days, 20 days, 10 days, or 5 days of data?

Average Daily Volume (ADV). Average daily volume computed over a historical period. We will use a rolling average in our forecast.

Median Daily Volume (MDV). Median daily volume computed over a historical period. We use a rolling median in our forecast.

Day of Week. A measure of the weekly cyclical patterns of trading volumes. Stocks tend to trade different percentages per days. This cyclical effect has varied over time and differs across market cap categories.

Lagged Daily Volume Term. We found some evidence of persistence in market volume. Many times both high and low volume can persist for days. However, the persistence is more often associated with high volume days due to the effect of trading large orders over multiple days to minimize price impact. Thus, when an institution is transacting a multi-day order, there is likely to be excess volume.

Author's Note: It is important to differentiate between the ADV measure used to normalize order size in the market impact estimate and the ADV or MDV measure used to predict daily volume. The ADV used in the former model needs to be consistent with the definition used by traders to quantify size. For example, if traders are using a 30-day ADV measure as a reference point for size, the market impact model should use the same metric. It is essential that the ADV measure that is used to quote order size by the trader be the exact measure that is used to calibrate the market impact parameters in the estimation stage. The daily volume forecast,